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An integrated relief network design model under uncertainty: A case of Iran

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ABSTRACT

In this study, a scenario-based robust approach is suggested for a multi-objective mixed-integer linear programming model in designing the relief network. A new approach to humanitarian inventory grouping problem based on the relief management objectives is proposed. The proposed model simultaneously optimizes inventory groups' number and corresponding service levels, assignment of relief commodities to groups, relief facility location, and relief service assignment. The proposed model aims to minimize the risk and the total cost of network management and simultaneously maximize the network population coverage. The fault tree analysis technique is used for vulnerability assessment of each demand point. To tackle the proposed optimization model, a hybrid Taguchi-based non-dominated sorting genetic algorithm-II is developed that incorporates an enhanced variable decomposition neighborhood search algorithm with fitness landscape analysis as its local search heuristic. The results illustrate the efficiency of the proposed model and solution algorithm in dealing with the considered disaster management issues.

1. Introduction

In recent years, the incidence of natural disasters and their costs of damages have steadily increased while the governments spend lots of money on disaster prevention and relief each year. Therefore, academic studies on disaster operations management and humanitarian logistics deserve particular attention. Disaster management aims to control the potential damages from hazards, guarantee prompt, appropriate support to victims of disasters, and achieve quick and effective recovery. The disaster management cycle demonstrates the ongoing process with the purpose of a strategy for lessening the impacts of disasters, responding through and directly after a disaster, and starting to recover after a disaster happened. Proper planning and actions at all stages of the disaster management cycle lead to more readiness, better warnings, vulnerability reduction or the prevention of disasters during the next implementation of the cycle. The complete disaster management cycle includes the determination of public strategies and tactics that either amends the origins of disasters or moderate their effects on the beneficiaries such as people, properties, and infrastructures (Wisner and Adams, 2002). There are four phases of disaster management include mitigation, preparedness, response, and recovery (McLoughlin, 1985). The mitigation involves steps to reduce the vulnerability to disaster impacts. The preparedness focuses on understanding how a disaster might impact the situation and how suitable preparation can build capacity to respond to needs and recover from a disaster. Immediate threats presented by the disaster and start of resources distribution are addressed by the response phase. Finally, the recovery phase is the restoration of all aspects of the disaster's impact on a community and the return of the condition to some sense of normalcy. An efficient design of relief chain networks plays an important role in disaster operation management. The proper relief network could be considered as an efficient connection between the preparedness and response phases (Bashiri and Hassanzadeh, 2016). In this paper, a comprehensive mathematical model for multi-objective strategic relief network design based on the concept of demand covering and hub location problem is proposed. The vulnerability of demand points under disaster is handled via using a fault tree analysis (FTA) method. An efficient hybrid meta-heuristic is proposed, which is based on a combination of the Taguchi-based non-dominated sorting genetic algorithm (NSGA)-II and particle swarm optimization. In addition, a fitness landscape analysis method is adopted to improve the systematic procedure of neighborhood structure selection in the proposed variable decomposition neighborhood search (VNDS) algorithm.

2. Literature review

Nowadays, disaster management is to prepare and plan for hazards in a proactive manner rather than waiting for them and reacting later.

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Therefore, many researchers have engaged in the disaster management (Galindo and Batta, 2013). A relief facility location problem is one of the most important issues in the disaster management (Caunhye et al., 2012a, 2012b). The facility location problem with reliability issue consideration has been well studied in the literature (Akgun et al., 2015). Various models based on mathematical programming for selection of facility location under disaster are proposed such as p-median (Aydin, 2016; Widener and Horner, 2011; Yushimito et al., 2012), p-center (Lu and Sheu, 2013; Lu, 2013), maximal covering problem (Jia et al., 2007; Balcik and Beamon, 2008; Doerner et al., 2009), and covering tour problem (Nolz et al., 2011). Various optimization models for disaster management such as a facility location, relief distribution, and casualty transportation have been studied in the previous studies (Akgun et al., 2015; Hasani and Mokhtari, 2018; Hasani and Zegordi, 2015b). KıLCı et al. (2015) proposed a mixed integer linear programming model for temporary shelter sites selection under earthquake occurrence in the Istanbul. Celik et al. (2016) proposed a two-stage stochastic mixed integer programming model for the decisions on facility location and pre-stocking levels for emergency supplies, and allocation of distribution centers. In addition, various proposed maximal covering location models with multiple coverage and facility location requirements combine the process of facility location with stock-prepositioning, evacuation, relief distribution, and vehicle routing model to perform relief distribution (Jia et al., 2007; Caunhye et al., 2012a). A joint location-inventory model is developed which tries to locate relief facilities, where they can cover and response the most demands of the vulnerable points (Belardo et al., 1984). Additional decision-making requirements such as a budget limitation and demand response time are considered by Balcik and Beamon (2008). Because of the importance of an inventory management in responding to demands of vulnerable points under disaster, relief distribution, and stock prepositioning mainly for a cost minimization issue are considered in some studies (Caunhye et al., 2012a). In another study, a priori planning instead of a post-disaster management is considered to acquire the more efficient emergency response in the event that it is needed via considering ordering costs, holding costs, and the possibility of post-disaster damage to stock (Rawls and Turnquist, 2010). Duran et al. (2011) propose a mathematical model for minimizing total response time to customers' demands under disaster. In another study, failures of supply facilities and linkage between facilities such as roadways damaged or destroyed in the relief distribution network are considered by using a scenario-based approach (Rawls and Turnquist, 2010). Ukkusuri and Yushimito (2008) proposed a location routing approach for the humanitarian prepositioning problem via considering the failures of facility capacity and linkage between facilities as well as a budget constraint. Campbell and Jones (2011b) developed a joint risk-based location-inventory mathematical model for prepositioning supplies in preparation for a disaster and determining an optimal stocking quantity at each relief facility.

As mentioned before, most of the conducted studies on the relief pre-positioning problem focused on determining the relief facility location and relief inventory management simultaneously with the aim of improving the quality of the relief operations (Caunhye et al., 2012a, 2012b). Relief inventory management is one of the key operations, which has an important impact on the relief management effectiveness. Beamon clarifies the general concept of stock repositioning in the humanitarian logistics, which is affected by various limitations such as budget, number of relief facilities, relief capacity, and uncertainty of relief demand. Akkihal (2004) proposed a mixed-integer mathematical programming model for simultaneously relief facility location and inventory pre-positioning for humanitarian operation with the aim of minimizing the total distance from the relief facilities to demand points. Hale and Moberg suggested that relief resources should be established in a manner as to not be exposed to risk. However, these facilities need to be closer to the disaster areas to which they are allocated to serve. Ozbay and Ozguven proposed a humanitarian inventory management model which can calculate the amount of the appropriate safety stock at a total minimum cost of relief inventory management. Balcik and Beamon (2008) develop a maximal covering location model for relief facility location and multi-commodity inventory management under budget and relief capacity constraints. Gatignon et al. assess the performance of the decentralized humanitarian supply chain of the international federation of the Red Cross operations under 2006 Yogyakarta earthquake in Indonesia. Lin et al. developed a mathematical model for disaster relief operations with the aim of minimization of total penalty cost of unsatisfied demand under natural disaster occurrence. Opit and Kim proposed a model for relief operation planning which determines the decision of relief facilities and amount of relief commodities should be kept at each relief facility. Tofighi et al. (2011) developed a relief network model based on the two-stage scenario-based stochastic programming approach for designing relief network under the disaster to determine an inventory pre-positioned level of relief commodities. Campbell and Jones (2011a) developed a risk-based mathematical model for determining the relief facility location and relief commodity inventory level at each relief facility. Ozguven and Ozbay proposed a multi-product stochastic humanitarian inventory management model for humanitarian emergency service providing to determine the optimal emergency inventory levels at the minimal cost. The interested readers are referred to Caunhye et al. (2012b) and Hoyos et al. (2015) for further information on the proposed optimization models for the location with relief distribution and stock pre-positioning problem in the literature.

In this study, a comprehensive scenario-based robust multi-objective mathematical model for relief network design subject to disruption is presented. While there are few studies on relief network design in literature, this study differs from the similar studies in terms of simultaneously considering triple objectives include minimizing demands serving risk; maximizing demand covering, and minimizing total relief network cost. Assessment of potential risk of relief network is handled via using the FTA method. To enhance the efficiency of inventory management decisions, a new approach for inventory grouping is incorporated with the decision of relief network management. Finally, an efficient solution algorithm based on the hybrid metaheuristic algorithm is proposed to obtain a list of Pareto optimal solutions. The proposed local search algorithm differs from the previous studies by using the customized neighborhood search structures via considering the fitness landscape analysis to improve the procedure of neighborhood structure selection. In addition, extensive computational experiments are conducted to investigate the performance of the proposed model as well as solution algorithm to solve considered instances.

3. Problem definition and formulation

In this paper, minimizing the maximum risk of deficiency in serving demand points by relief centers under disaster is studied. Locating the relief facilities near the disaster areas has an important impact on minimizing a response time. However, this decision would be risky because the established facilities near the vulnerable points could be disrupted. Therefore, the established relief facility may not support the demand points properly as a consequence of the severity of disaster impact on the infrastructure. In the domain of the risk analysis, three main components of risk are considered in this study based on the alleged real-life experiences include threat, vulnerability, and consequence (Willis et al., 2005). The threat is the probability of disaster occurs during a specified period. The vulnerability is defined as the probability that the damages occur. Compensations take place at a demand point because it is not served on time by the located facilities during a specified period. In our case, the vulnerability is calculated via using the FTA method. A consequence, as an outcome of the damage, has a specific effect on the targets such as the number of population living in a disaster zone. Finally, the expected risk is computed as the product of threat, vulnerability, and consequence.

At the first step of performing the FTA method, disaster condition and related top-down disruption are defined based on the past experiences and experts' opinions. Expert team in this study constitutes with experts from the Iranian Red Crescent Society (RCS) and the National Disaster Management Organization (NDMO). Then, experts' opinions are used to determine the possible reasons for the considered disruptions. These data are presented in the level just below the top-level disruption in the fault tree. Each element is broken down with additional gates to related lower levels. The gate selection is based on the relationships between the considered elements. The fault tree is completed if all the chains are terminated in a basic disruption. The probability of each occurrence is evaluated for the lowest level elements from the bottom up.

In our case, it is assumed that all the relief facilities could be at all the vulnerable nodes which are considered as demand points. To compute the vulnerability of each demand point, the fault tree with a top event "Target" is recognized as the event that the considered demand point is not served by the related facilities. The top event may happen if none of the relief facilities assigned to the considered demand point can support properly. The location of each relief facility is not known a priori. Therefore, an input event is defined for each potential facility location and linked to the top event with an "AND" gate. The other branches of the fault tree are computed in the same way. We define the event that the facility at one node cannot support the other demand points and hence three potential events are identified which are connected to the input event via an "OR" gate of these three events: (E1) A relief facility is not located at the desired point, (E2) A facility is located at the desired point, but does not cover the considered demand point based on a predetermined coverage distance, and (E3) A facility is located at a desired point and the demand point is covered; however, the facility is disrupted and inaccessible. The second event could be broken down into two more base events with an "AND" gate: (*E21*) A facility is located at the desired point, and (*E22*) A facility at the desired node does not cover the considered demand point. In addition, the third event could be broken down into three events via an "AND" gate: (*E31*) A facility is located at the desired node, (*E32*) A facility at the desired node covers the considered demand point, and (*E33*) A facility at the desired node is disrupted and hence inaccessible. Of these three events, the first two ones are the base events. There may be several events leading to the disruption of the facility at the desired node. In our case, four events are defined which cause disruptions include an earthquake, flood, landslide, avalanche. These four events are connected to its upper event via using an "OR" gate. The aforementioned top-down approach based on the considered tree structure is repeated for all the demand points. Then, the vulnerability of each demand point is computed. It should be noted that the locations of the relief facilities are not already known, but the vulnerability of a demand point is directly dependent on where the relief facilities are established. Therefore, the probability of a top event cannot be calculated directly by using probabilities; the facility location decisions are to be integrated into the vulnerability computation (see Eq. (4)). Then, the uncertainty of the level of disaster occurrence includes the severity and extent of the disaster, as well as its impact on the planning parameters such as relief demand and capacity of relief facility, are handled by the scenario-based robust optimization. A hierarchical structure of the relief network between relief centers and higher-level centers with different coverage radius is considered.

To enhance the efficiency of the inventory system management, there is a need to classify the inventory items into specific groups and then to adopt an appropriate inventory policy for each considered group. The inventory policy could include some key decisions such as determine an appropriate service level to satisfy customers' demands. The proper service level shows a trade-off between the costs of inventory holding and stock-outs. Determining this equilibrium condition is an important issue for managing relief service providing under disaster occurrence. There is a budget constraint for managing relief network for pre-disaster and post-disaster planning. The uncertainty of disaster occurrence and its impact on the relief demand and service capacity are handled by the scenario-based stochastic approach. Each scenario has a specific occurrence probability. Under each scenario, it is assumed that the relief demand has a normal distribution, with a specific mean and standard deviation, as well as a constant lead time. The proper allocation of inventory items to considered inventory groups has a considerable impact on the inventory system service level and total cost. Therefore, the costs of maintaining and managing inventories groups are considered. In the proposed model, the tradeoff between two main performance measures include service level, correlated with a number of groups, and inventory system cost is considered. Each relief center is located in a single disaster area and can provide relief service for vulnerable points. In the classical ABC method, only three inventory groups are considered because of the simplicity in implementation and low administrative cost of maintaining. The cost of maintaining and managing inventory group is not constant and it is a decreasing function of the number of groups because of the economies of scale. In the proposed optimization model, various strategic and tactical decisions are regarded such as establishing a relief center at a vulnerable point, selecting the number of inventory groups at each relief center, and assigning each inventory item to an appropriate group. The aim of the proposed model is to minimize the risk of providing service, minimize the total cost of relief network, and maximize the total network coverage simultaneously. In this problem, facility capacity limits and multiple radiuses of service coverage are considered. In addition, costs of relief facility establishment, material shipment, as well as not satisfying demand are considered.

3.1. Mathematical model

In this section, a comprehensive mathematical model for the relief network design problem is developed.

Sets:

V	Set of vulnerable points (i.e., $I \subseteq V$, $J \subseteq V$),
R	Set of relief centers (i.e., $K \subseteq R$, $L \subseteq R$),
R^v	Set of relief centers that provides service in disaster area v,
P	Set of coverage radius,
A	Set of inventory items,
E	Set of inventory groups,
M	Maximum number of inventory groups,
T	Set of planning periods,
S	Set of scenarios.

Parameters:

C^1_{iklj}	Linkage cost between vulnerable points i and j, which points i and j are allocated to higher level centers k and l, respectively.
C^2_{ika}	Transportation cost of item a between vulnerable point i and relief center k,
Sc^1_{ia}	Cost of not satisfying the demand of inventory item a at vulnerable point i,
Sc^2_{ik}	Cost of capacity shortage at relief center k at vulnerable point i,
$Rc_{ia(s)t}$	Demand of inventory item a at vulnerable point i (under scenario s) in period t,
Cr_{kp} and Ch_{kp}	Covering radius p of the relief center k and higher-level center k, respectively,
D_{ik}	Distance between vulnerable points i and k,
F^1_{ki} and F^2_{ki}	Fixed cost of establishing the higher level center and the relief center k at vulnerable point i, respectively,
$Cap_{k(s)}$	Capacity of relief center k (under scenario s),
$Lcap_{k(s)}$	Capacity loss of relief center k (under scenario s),
$T_{i(s)}$	Probability that a disruption occurs at demand point i (under scenario s),
Prb_s	Probability that a scenario s occurs,

E_i	Vulnerability of the demand point i results only from the disruption of the facility for some reasons,
$\delta_{ia(s)}$	Standard deviation of demand of item a at vulnerable point i (under scenario s),
$L_{ia(s)}$	Lead-time of supplying item a at vulnerable point i (under scenario s),
C_{ak}	Holding cost of inventory item a at relief center k,
AC_{ka}	Unit holding cost of inventory item a at relief center k,
ND_{ka}	Unit unsatisfying demand cost of inventory item a at relief center k,
CW_a	Critically weight of inventory item a,
M	A very large positive number,
N	Total number of inventory items,
ω_e	Overhead management cost of inventory group e,
B_0	Pre-disaster budget,
B_1	Post-disaster budget,
$T_{iak(s)}$	Expected time to respond relief demand of item a at vulnerable point i from relief center k (under scenario s),
$RT_{i(s)}$	Maximum response time limit to perform emergency response in vulnerable point i (under scenario s),
MRS_{ia}	Maximum response time limit to perform emergency response to demand of item a at vulnerable point i,
UD_{ai}	Upper bound on the proportion of unsatisfied demand of inventory item a at vulnerable point i,
M_a	Degree of importance of inventory item a,
Vo_a	Unit volume of inventory item a,
Z_a	The upper bound of the proportion of unsatisfied relief demand of item type a,
λ	Wight devoted to variability part of the second objective function.

Variables:

y_{ikpt}	1 if a relief center k is established at vulnerable point i with a coverage radius p in period t; 0 otherwise,
x_{ikt}	1 if vulnerable point i is allocated to higher-level center k in period t,
a_{ikt}	1 if vulnerable point i is covered by relief center k in period t,
x_{ikljt}	1 if vulnerable point i is connected to j through higher-level centers k and l in period t; 0 otherwise,
$mf_{ika(s)t}$	Flow of inventory item a between vulnerable point i and its allocated higher-level center k (under scenario s) in period t,
θ_s	Linearity coefficient for the second objective function,
α_e	Service level associated with group e,
z_e	Z-value associated with the service level α_e of group e,
y_{ek}	1 if inventory group e is selected at relief center k, and 0 otherwise,
x_{aek}	If inventory item a is assigned to group e at relief center k,
v_{kat}	Inventory level of item a at relief center k in period t,
$f_{ika(s)t}$	The proportion of item type a relief demand satisfied by relief center k that provide services in vulnerable point i (under scenario s) in period t,
$n_{ia(s)t}$	The proportion of unsatisfied relief demand of item type a at vulnerable point i (under scenario s) in period t,
a_{ik}^1	A set of the potential response time of the relief center k that will provide service at vulnerable point i ($a_{ik} = 1$; if expected time T is no greater than the maximum response time limit, 0 otherwise),

<u>**Deterministic Model:**</u>

$$Max\quad f_1 = \sum_{i,a,k,t} Rc_{iat}.\ a_{ikt} \tag{1}$$

$$Min\quad f_2 = \sum_{i,j,k,l,t} C^1_{iklj}.\ x_{ikljt} + \sum_{k,i,t} F^1_{ki}.\ x_{ikt} + \sum_{i,r,k,t} F^2_{ki}.\ y_{ikpt} + \sum_{i,k,a,t} C^2_{ika}.\ mf_{ikat} + \sum_{i,r,a,t} Rc_{iat}.\ Sc^1_{ia}.\ n_{iat} \tag{2}$$

$$Min\quad f_3 = Mr \tag{3}$$

$$Mr \geqslant \prod_{i,k,l,j,p} Sc^2_{ik}\cdot T_{iak}\cdot((1-y_{ikpt}) + ((1-a_{ikt})\cdot x_{ikljt}) + E_i a_{ikt} y_{ikpt})\quad \forall\ t \tag{4}$$

$$x_{ikljt} \leqslant x_{ikt}\quad \forall\ i,\ k,\ l,\ j,\ t \tag{5}$$

$$x_{ikljt} \leqslant x_{jlt}\quad \forall\ i,\ k,\ l,\ j,\ t \tag{6}$$

$$x_{ikt} \leqslant y_{ikpt}\quad \forall\ i,\ k,\ t \tag{7}$$

$$\sum_{k,l} x_{ikljt} = y_{ikpt}\cdot y_{jlpt}\quad \forall\ i,\ j,\ r,\ t \tag{8}$$

$$\sum_{i,a} Vo_a\cdot mf_{ikat} + Vo_a\cdot v_{ka,t-1} \leqslant Cap_k\cdot y_{ikpt}\quad \forall\ k,\ t \tag{9}$$

$$\sum_{k} mf_{ikat} \geqslant Rc_{iat}\cdot \sum_{k,p} y_{ikpt}\quad \forall\ i,\ a,\ t \tag{10}$$

$$\sum_{k,r} a_{ikt}\cdot y_{ikpt} \geqslant 1\quad \forall\ i,\ t,\ p \tag{11}$$

$$D_{ik}\cdot a_{ikt} \leqslant Cr_{kp}\cdot y_{ikpt}\quad \forall\ i,\ k,\ t \tag{12}$$

$$D_{ik}.\ x_{ikt} \leqslant Ch_{kp}\cdot y_{ikpt}\quad \forall\ i,\ k,\ p,\ t \tag{13}$$

$$\sum_{k} f_{ikat} = 1-n_{iat}\quad \forall\ i,\ a,\ t \tag{14}$$

$$\sum_{i} n_{iat} = \sum_{e,i,k} z_a \sigma_{iat} \sqrt{L_{ia}}\, x_{aek}\quad \forall\ a \tag{15}$$

$$n_{iat} \leqslant z_a\quad \forall\ i,\ a,\ t \tag{16}$$

$$\sum_{i} mf_{ikat} + v_{ka,t-1} - \sum_{i} Rc_{iat}.\ f_{ikat} = v_{kat}\quad \forall\ k,\ a,\ t \tag{17}$$

$$\sum_{k,i,t} F^1_{ki}\cdot x_{ikt} + \sum_{i,r,k,t} F^2_{ki}\cdot y_{ikpt} + \sum_{k,a,t} Ac_{ka}\cdot v_{kat} \leqslant B_0 \tag{18}$$

$$\sum_{i,k,a,t} Rc_{iat}\cdot Sc^2_{ik}\cdot n_{iat} + \sum_{i,k,t} C^2_{ika}\cdot mf_{ikat} \leqslant B_1 \tag{19}$$

$$T_{iak}\cdot a_{ikt} \leqslant \delta_{ka}\quad \forall\ i,\ k,\ a,\ t \tag{20}$$

$$a_{ikt} \geqslant x_{ikt}\quad \forall\ i,\ k,\ t \tag{21}$$

$$\sum_{i,k} x_{ikt} \geqslant 1\quad \forall\ t \tag{22}$$

$$\sum_{k} x_{ikt} \leqslant 1\quad \forall\ i,\ t \tag{23}$$

$$\sum_{a} x_{aek} \leqslant N\cdot y_{ek}\quad \forall\ e,\ k \tag{24}$$

$$v_{kat} = \sum_{e} Rc_{ia}\cdot L_{ia}\cdot x_{aek} + \sum_{e} z_a\cdot \sigma_{ia}\cdot \sqrt{L_{ia}}\cdot x_{aek}\quad \forall\ a,\ k,\ i,\ t \tag{25}$$

The objective function (1) tries to maximize the total coverage by the relief network within the range of the desired service distance. The objective function (2) tries to minimize the total cost of relief network design include the cost of relief facilities linkage, the fixed cost of facility establishment, the cost of material shipment, and shortage costs. The objective function (3) tries to minimize the maximum risk of total demand nodes. Constraint (4) shows the maximum risk of vulnerable points based on the FTA risk assessment method. Constraints (5)–(7)

show the assignment of each vulnerable point to another vulnerable point which has been selected as a higher-level center. Constraint (8) ensures that the path between two vulnerable points should pass at least one higher-level center. Constraint (9) ensures the capacity limit for each relief center to satisfy demands of multiple points. Constraint (10) ensures that all the vulnerable points should be covered by the established relief facilities. Constraint (11) ensures that each relief center could be allocated to one higher-level center via considering the covering radius of the higher-level center. Constraints (12) and (13) demonstrate the maximum distance coverage of each relief center and higher relief center, respectively. Constraint set (14) demonstrates the service providing to satisfy relief demands of vulnerable points by each established relief center under disaster. Constraints set (15) and (16) shows the amount of satisfied and unsatisfied relief demands of each demand point in vulnerable areas. Constraint (16) shows the maximum acceptable level of unsatisfied demand of each inventory item k. Constraint (17) demonstrates the total inventory level of each item at each relief center to satisfy the demand points. Constraint (18) ensures that the total cost of relief center preparation to provide service should be less than the available budget of pre-disaster management. Constraint set (19) quarantines that the total cost of service providing under disaster should be less than the available budget for post-disaster management. Constraints (20) and (21) indicate that the each established relief center only provides relief services for demand points which are in the considered response time region. Constraint (22) guarantees that each demand point under disaster should be supplied by at least one relief center. Constraint set (23) indicates that each inventory item in each relief center is assigned to only one inventory group. Constraint set (24) enforces that inventory items can be assigned only to selected inventory groups. Constraint set (25) calculates the inventory level of each relief item.

Robust Model:

The robust counterpart of the model (1)–(25) is developed based on the by hypothesis proposed by Yu and Li (2000). The deterministic model is changed as below to form the robust counterpart by using new Eqs. (27), (29), and (30) and the defined sets and notations. Eq. (27) considers a mean value of different scenarios, mean absolute deviations, and infeasibility penalties for not satisfying the demands of vulnerable points under disaster which make the model infeasible. Eq. (28) computes the total cost of the relief network management for each scenario. Eq. (30) considers the penalty of the objective function of the robust model and is used for conditions where some of the considered scenarios are infeasible. Eqs. (36) and (37) are counterparts of Eqs. (9) and (10), respectively. It should be noted that nonlinear terms of the proposed model such as the terms of the product of two binary variables) are linearized.

$$Max \quad f_1 = \sum_{i,j} \Pr b_s \cdot Rc_{iast} \cdot a_{ikt} \tag{26}$$

$$Min \quad f_2 = \left(\begin{array}{l} \sum_s prb_s \cdot \Gamma_s + \lambda \sum_s prb_s \left[\left(\Gamma_s - \sum_s prb_s \cdot \Gamma_s \right) + 2\theta_s \right] \\ + \sum_{i,k,a,t,s} \Pr ob_s \cdot Sc_{ia} \cdot n_{iast} + \sum_{i,k,t,s} \Pr ob_s \cdot Sc_{ki} \cdot lcap_{kst} \end{array} \right) \tag{27}$$

$$Min \quad f_3 = Mr \tag{28}$$

$$\Gamma_s = \sum_{i,j,k,l,t} C^1_{iklj} \cdot x_{ikljt} + \sum_{k,i,t} F^1_{ki} \cdot x_{ikt} + \sum_{i,k,p,t} F^2_{ki} \cdot y_{ikpt} + \sum_{i,k,t} C^2_{ika} \cdot mf_{ikast} + \sum_{i,r,a,t} Rc_{iast} \cdot Sc_{kai} \cdot n_{iat} \tag{29}$$

$$\left(\Gamma_s - \sum_s prb_s \cdot \Gamma_s \right) + \theta_s \geqslant 0 \quad \forall\ s \tag{30}$$

$$Mr \geqslant \prod_{i,k,l,j,p} Sc^2_{ik} \cdot T_i \cdot ((1 - y_{ikpt}) + ((1 - a_{ikt}) \cdot x_{ikljt}) + E_i a_{ikt} y_{ikpt}) \quad \forall\ t \tag{31}$$

$$x_{ikljt} \leqslant x_{ikt} \quad \forall\ i,\ k,\ l,\ j,\ t \tag{32}$$

$$x_{ikljt} \leqslant x_{jlt} \quad \forall\ i,\ k,\ l,\ j,\ t \tag{33}$$

$$x_{ikt} \leqslant y_{ikpt} \quad \forall\ i,\ k,\ p,\ t \tag{34}$$

$$\sum_{k,l} x_{ikljt} = y_{ikpt} \cdot y_{jlpt} \quad \forall\ i,\ j,\ k,\ l,\ p,\ t \tag{35}$$

$$\sum_{i,a} Vo_a \cdot mf_{ikast} + \sum_a Vo_a \cdot v_{kas,t-1} = Cap_{ks} \cdot y_{ikpt} + Lcap_{ks} \quad \forall\ i,\ k,\ p,\ t,\ s \tag{36}$$

$$\sum_k mf_{ikast} + n_{iast} = Rc_{iast} \cdot \sum_{k,p} y_{ikpt} \quad \forall\ i,\ s,\ p,\ t \tag{37}$$

$$\sum_{k,p} a_{ikt} \cdot y_{ikpt} = 1 \quad \forall\ i,\ t \tag{38}$$

$$D_{ik} \cdot a_{ikt} = Cr_{kp} \cdot y_{ikpt} \quad \forall\ i,\ k,\ t \tag{39}$$

$$D_{ik} \cdot x_{ikt} = Ch_{kp} \cdot y_{ikpt} \quad \forall\ i,\ k,\ t \tag{40}$$

$$\sum_j mf_{ikast} = 1 - n_{iast} \quad \forall\ i,\ a,\ s,\ t \tag{41}$$

$$\sum_i n_{iast} = \sum_{e,i} z_a \sigma_{ias} \sqrt{L_{ias}} x_{aek} \quad \forall\ a,_s,\ t \tag{42}$$

$$n_{iast} \leqslant z_a \quad \forall\ i,\ a,\ s,\ t \tag{43}$$

$$\sum_i mf_{ikast} + v_{kas,t-1} - \sum_i Rc_{iast} \cdot mf_{ikast} = v_{kast} \quad \forall\ k,\ a,\ s,\ t \tag{44}$$

$$\sum_{k,i,t} F^1_{ki} \cdot x_{ikt} + \sum_{i,r,k,t} F^2_{ki} \cdot y_{ikpt} + \sum_{k,a,t} Ac_{ka} \cdot v_{kat} \leqslant B_0 \tag{45}$$

$$\sum_{i,k,a,t} Rc_{iast} \cdot Sc^2_{ik} \cdot n_{iast} + \sum_{i,k,t} C^2_{ika} \cdot mf_{ikast} \leqslant B_1 \quad \forall\ s \tag{46}$$

$$T_{iak} \cdot a_{ikt} \leqslant \delta_{ka} \quad \forall\ i,\ a,\ k,\ t \tag{47}$$

$$a_{ikt} \geqslant x_{ikt} \quad \forall\ i,\ k,\ t \tag{48}$$

$$\sum_{i,k} x_{ikt} \geqslant 1 \quad \forall\ t \tag{49}$$

$$\sum_k x_{ikt} \leqslant 1 \quad \forall\ i,\ t \tag{50}$$

$$\sum_a x_{aek} \leqslant N \cdot y_{ek} \quad \forall\ e,\ k \tag{51}$$

$$v_{akt} = \sum_e Rc_{ias} \cdot L_{ias} \cdot x_{aek} + \sum_e z_a \cdot \sigma_{ias} \cdot \sqrt{L_{ias}} \cdot x_{aek} \quad \forall\ a,\ k,\ i,\ t \tag{52}$$

4. Solution algorithm

Due to the NP-hard nature of the supply chain network design problem, solving large-size instances efficiently within a reasonable time is a challenging task (Fahimnia et al., 2013; Dixit et al., 2016). The outcomes of applying the genetic algorithm as a most popular evolutionary algorithm to a variety of single as well as multi-objective network design problems have been very promising among the other evolutionary algorithms (Altiparmak et al., 2008; Fahimnia et al., 2013). In spite of strengths of the genetic algorithm, it has a very limited effect in intensifying the local search process (Hasani and Zegordi, 2015a). To tackle this challenge, a hybrid genetic algorithm such as a memetic algorithm has been proposed (Moscato and Norman, 1992). In this study, an efficient hybrid meta-heuristic based on the genetic algorithm is proposed that incorporates the Taguchi-based NSGA-II, particle swarm optimization, and the VNDS algorithm. In the proposed solution algorithm, solution space search is handled by the Taguchi-based NSGA-II. The Pareto improvement is obtained at each

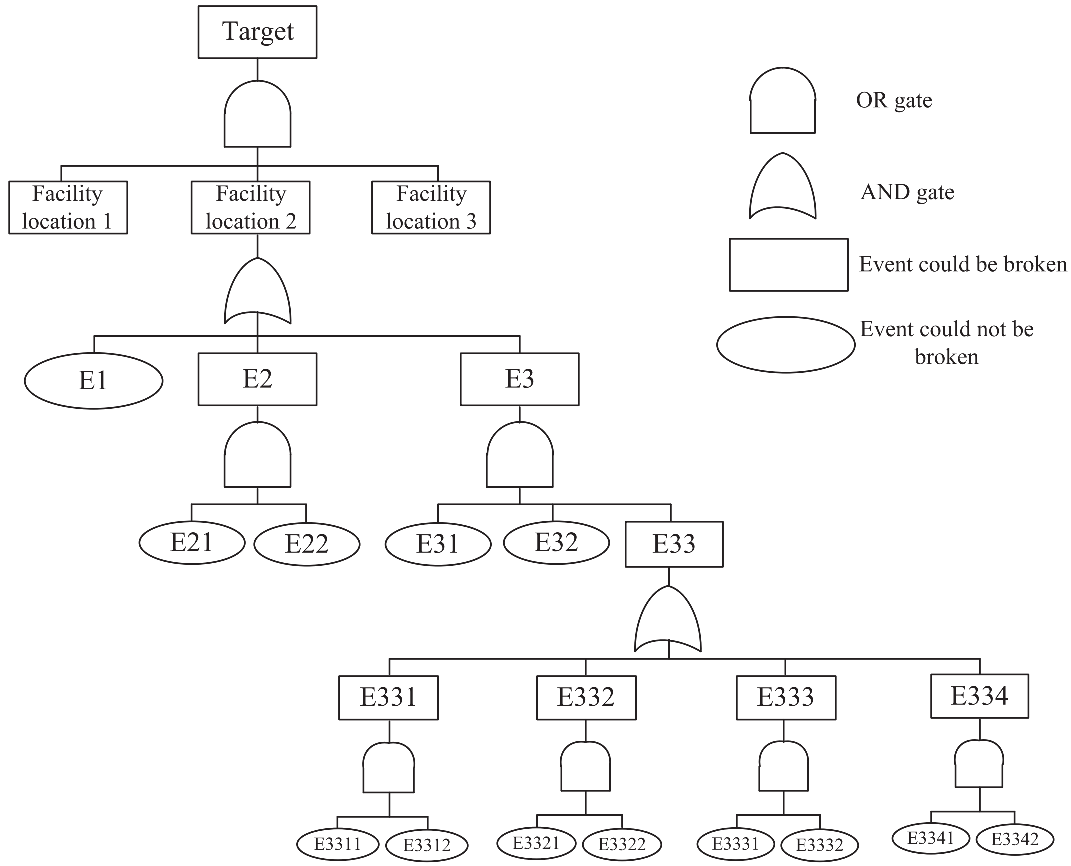

Fig. 1. The schematic overview of the FTA for three potential facility locations.

generation of the proposed NSGA-II via using the customized operators inspired from the particle swarm optimization and the VNDS. The proposed VNDS algorithm investigates the solution space via new neighborhood structures. A fitness landscape analysis is applied to improve the systematic procedure of neighborhood structure selection. In addition, a hill climbing heuristic is used within the proposed VNDS to more intensify a search. The schematic structure of the proposed hybrid solution algorithm, namely HTNSG-II-VNDS, is presented in Fig. 1.

4.1. Encoding and fitness evaluation of the solution chromosomes

Solution chromosome of the proposed hybrid solution algorithm is represented via using a two-dimensional binary array. The size of the proposed binary array is $(|A.E.J| + |E.J| + |I| + |I.K| + |I.J| + |I.J.K.L|).|T|$. Fitness evaluation of each solution chromosome is based on the values of the three objective functions. Via using values of the solution chromosome's genes, the binary decision variables are generated. Then, the mixed integer linear problem transforms into a single objected linear programming one which is solved by the LINDOGLOBAL in GAMS 24.1.2 optimization software. An adaptive weighted-sum method is used for transforming the multi-objective optimization model into a single objective one (Kim and de Weck, 2006).

4.2. Initial and next population generation

The initial population of the proposed hybrid solution algorithm is generated randomly. The next populations are generated in three ways. In the first method, a predefined number of solutions are generated via the Taguchi-based crossover (Yang et al., 2013). In the second method, a multi-point mutation operator is used to generate a specific number of solutions. The employed multi-point mutation operator flips the binary value of the selected gene randomly with a probability of 0.5. Genes in each segment of the chromosome are selected randomly with a probability of 0.5. Finally, a customized operator of the particle swarm optimization is utilized to improve two Pareto optimal solution sets, including Pareto optimal solutions of each generation and overall generations. This operator is a kind of a multi-point crossover, which is applied to decision variables as follows: (1) if both genes of selected parents are equal to one or zero, then a related gene of a child has an equal value to its related genes of the parents; otherwise, the binary value of the child's gene is determined randomly. A roulette wheel selection scheme is considered to select the fittest individual prior to others into the next generation via considering selection probability in portion of its fitness values. Selected individuals from the mating pool could be used for applying the Taguchi-based crossover, multi-point mutation and particle swarm optimization operators with predetermined probabilities (Hasani and Khosrojerdi, 2016).

4.3. The proposed variable neighborhood decomposition search algorithm

Hansen and Mladenovic (2001) proposed variable neighborhood search based on the idea of successive search a set of predefined neighborhoods to obtain a better solution. This algorithm searches systematically a set of neighborhoods to develop diverse local optimum and to move well away from the local optimum. Via using approximation decomposition method, an efficiency of the variable neighborhood search will be enhanced for solving large instances. The sequence of subproblems of the main problem with smaller sizes than the initial one is generated from the different preselected set of the neighborhoods. Complementary of the selected neighborhood structures during a search process is the main requirement to have an effective solution space search with the VNDS. In this study, a fitness landscape approach is used to determine the most effective order of

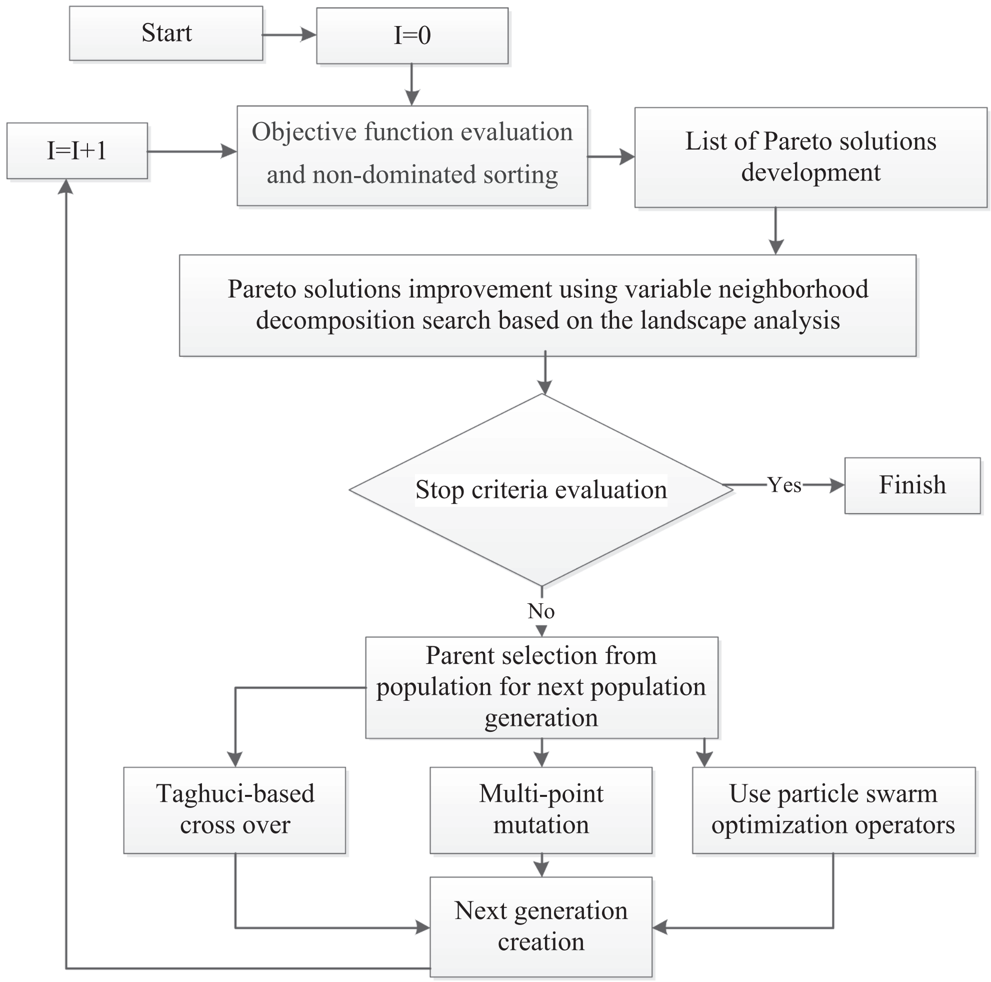

Fig. 2. The schematic overview the proposed hybrid solution algorithm structure.

neighborhood selection. This approach could realize that who and where neighborhood structures work efficiently. To analyze the landscape of the proposed problem, the normalized average distance is adopted for population *p*. Due to binary representation in the proposed search space, the Hamming distance (see Eq. (53)) is used Talbi (2009).

$$Dmm(P) = \sum_{s\in P, t\in P, t\neq s, i=1} s_i \oplus t_i \Big/ |P|\cdot(|P|-1)\cdot\left(\max_{s,t\in P} dist(s,t)\right) \tag{53}$$

Steps of the proposed variable neighborhood decomposition search are demonstrated in Fig. 2. We make use of the advantage of the VNDS and apply different neighborhood structures for the network design problem which are listed in the following.

k1: A single random pairwise interchange in which two departments are selected randomly and their positions are swapped.
k2: Multiple pairwise interchanges in which for a specific number of iterations departments are selected randomly and their positions are exchanged.
k3: A subsequence moving operator in which a group of departments is moved to another position altogether.
k4: An insertion mechanism in which a randomly selected department is omitted from its position and inserted between two other positions selected randomly.
k5: A subsequence shuffling operator in which a group of departments is selected and jumbled up. In other words, they are positioned without any specific order.
k6: An inversion structure in which a group of departments is selected and positioned inversely.
k7: A subsequence moving and inversion operator in which a group of departments is positioned inversely and moved to another position altogether.
k8: A shifting neighborhood structure in which two random positions are selected depends on the first position, which is greater or less than the other one, the department in the position of the first random number is shifted backward or forward, respectively.
k9: An adjacent swap operator in which a department is selected randomly replaces its position probabilistically with its left or right position.
k10:

4.4. Transferring Pareto solutions to the next generation

The remaining of the population is generated via transferring predetermined percent of the Pareto solutions from the previous generation to the next one. A pre-determined number of individuals of the first non-dominated front transfer from the previous generation to the next generation. These individuals are selected via considering Pareto front

Initialization. Select the set of neighborhood structures $N_k, k = 1, ..., k_{max}$, that will be used in the search; find an initial solution x; choose a stopping condition; Repeat the following sequence until the stopping condition is met:

(1) Set $k \leftarrow 1$;

(2) Until $k = k_{max}$, repeat the folowing steps:

(*a*) Shaking. Generate a point x′ at random from the k^{th} neighborhood of x $(x' \in N_k(x))$; in other words, let y be a set of k solution attribute present in x' but not in x $(y = x' \setminus x)$.

(*b*) *Local search*. Find the best solution in the space of y either by inspection or by some heuristic; denote the best solution found with y′ and x″ the corresponding solution in the whole space S $(x'' = (x' \setminus y) \cup y')$

(*c*) *Move or not*. If the solution thus obtained is better than the incumbent, move there $(x \leftarrow x'')$, and continue the search with N_1 $(k \leftarrow 1)$; otherwise, set $k \leftarrow k+1$;

Fig. 3. The pseudo-code of the proposed VNDS heuristic.

rank and crowded distance measure. A crowded distance (D) is measured by $D = \sum_j \frac{|F_j^{i+1} - F_j^{i-1}|}{|F_j^{\max} - F_j^{\min}|}$ (Deb et al., 2002).

4.5. Non-dominated sorting procedures and Pareto list updating

Members of the population are sorted based on the ranking scheme has been proposed by Souza et al. (2010). This efficient runtime sorting algorithm is represented in Fig. 3. In each generation of the NSGA-II, Pareto list is updated regularly. Members of each population generation in the first non-dominated front are selected for Pareto list generation. (see Fig. 4)

5. Computational study

The proposed planning model described in Section 3 is implemented through a computational study using data from the study on a seismic micro-zoning of the greater Tehran area in the Islamic Republic of Iran. The obtained results provide insight into the effects of considering risk in robust multi-objective relief network design decision-making on efficiency and effectiveness.

5.1. Context for study

A significant number of Iran cities are at a high risk of exposure to the severe earthquake (Pacific Consultants International, 2000). Among

Faster_non_dominated_sort (P)

{

For each objective i

S_i = sort_on_objective (P, i) a

For each p in P

$$P_{rank} = \sum_{i=1}^{M} position(p, S_i)$$

}

Sort_on_objective (P, i)

{

$Pos = 0$

Q_i = sort (P)

For each p_j in P

If $(p_j(obj) \neq p_{j-1}(obj))$

$Pos = pos + 1$

Set_position (p_j, Q_i, pos)

Return Q_i

}

Note: S_i represents an array of population individual sorted by each objective i, position of p in S_i is displayed by $position(p, Si)$, standard arrangement of P depended on objective value that is represented by $sort(P)$, and objective value of population individual P_j is displayed by $p_j(obj)$. Finally, theposition of p_j as pos in Q_i is set by $set\_position(p_j, Q_i, pos)$.

Fig. 4. The pseudo code of the non-dominated sorting algorithm.

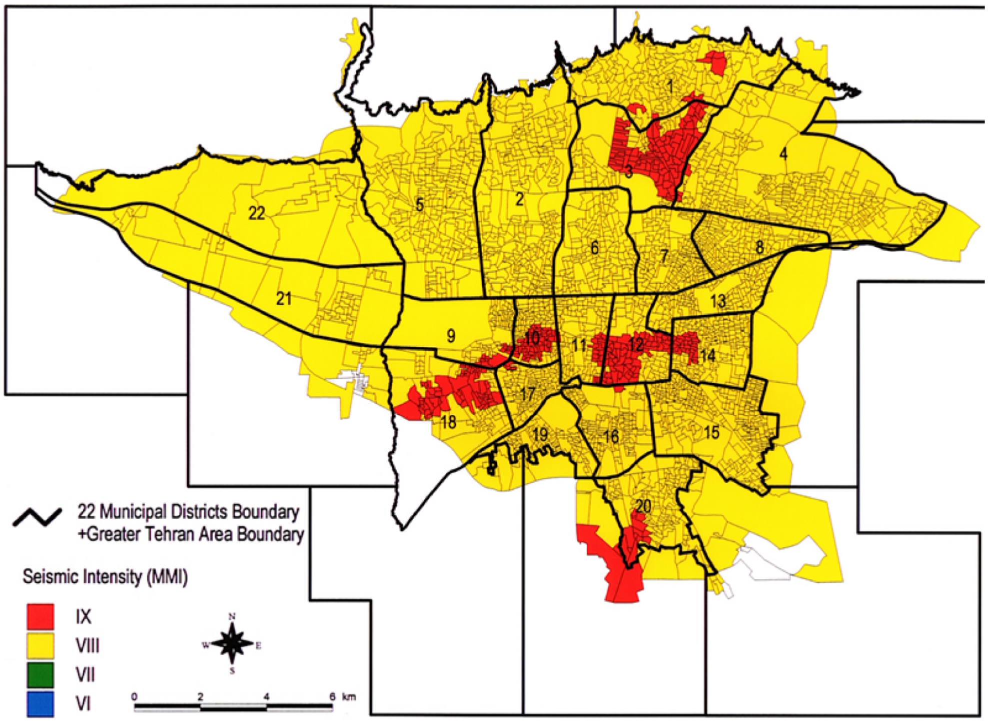

Fig. 5. The seismic intensity distribution map of Tehran (Pacific Consultants International, 2000).

of these high-risk cities, conditions of the Tehran and its surrounding areas are most critical. Tehran is the economic and political center of Iran. About 30 percent of Iran's public-sector workforce and 45 percent of its large industrial companies are located in Tehran. Historical seismic data indicate that Tehran has severely suffered from numerous severe earthquakes with recurrence periods of 150 years. Earthquake researchers have confidence in a severe earthquake will strike the Tehran area in the near future because the Tehran has not experienced a strong earthquake since 1830. It is directly essential to develop a comprehensive earthquake disaster prevention plan, considering regional and urban requirements to diminish possible earthquake damages in Tehran. An integrated relief center network design could have a significant impact on the quality of relief service providing under disaster. As shown in Fig. 5, the earthquake potential intensity in the northern area of the Tehran city would be 9 Richter and in the southern area would be 7 Richter. The most area of the Tehran city experiences an earthquake intensity of 8 Richter. The total number of damaged constructions is predicted as 480,000 in Tehran. The total estimated destruction ratio is about 55 percent. It should be mentioned that the number of destructed constructions in District 15 is the largest. The total number of the constructions is also the largest in district 15. The destruction ratio in Districts 11, 12, 16 to 20 is at a high value of about 80 percent. One of the main reasons for this high construction ratio is the presence of numerous in danger structures and the severe possible seismic motion, MMI = 9, in these districts. The ratio of earthquake damage in Districts 1 to 5, which are positioned in the northern part of the Tehran city, is relatively small. Therefore, there is a need to develop an integrated and a comprehensive plan for relief facility locations to provide an appropriate service level under disaster.

5.2. Study design

For the sake of data confidentiality, scaled limits of the parameters of the proposed mathematical model in the considered case study are presented in Table 1. Five test problems are considered based on the comments of experts on the disaster management in Iran (see Table 2). Comprehensive factors are considered for potential relief facility locations such as earth structure, street network, staying away from danger zones, closing to rescue centers, closing to population concentration centers, accessing to appropriate space and infrastructures to provide relief services properly.

5.3. Computational results

The effectiveness of the proposed solution algorithm is investigated using a series of computational experiments. The proposed solution algorithms and the mathematical model are coded in *C#* and GAMS 24.2.2, respectively.

5.3.1. Performance measures

In this study, two main criteria, including runtime and solution quality, are considered to evaluate the performance of the considered optimization algorithms. Various measures are proposed in the literature to assess the solution quality of multi-objective algorithms, which can be categorized into two main groups: (1) Diversity metrics such as hyper-volume and set coverage (Zitzler and Thiele, 1999), (2): Convergence metrics such as spacing metric (Deb et al., 2001). Assessment of the solution quality for the multi-objective NP-hard problem is more difficult due to inability to describe the quality of the approximation in terms of a number of criteria, such as closeness to the Pareto set (Baños et al., 2013). To tackle this problem in this study, a quasi-optimal Pareto front as a reference set is created via combing the obtained solution by the five considered algorithms and maintaining the non-dominated ones. Because of importance of both the convergence and the diversity metrics (Baños et al., 2013, Deb et al., 2002), three metrics including set coverage, namely *Setc*, hyper-volume, namely *HV*, and spacing metric, namely *SM*, are considered to evaluate the quality of the obtained Pareto sets in this study as follows:

- *HV* metric: The *HV* calculates the area of the objective space, which is bounded by a reference point. Therefore, a Pareto set with a larger value of *HV* is expected to present a better set of non-dominated solutions.
- *Setc* metric: The *Setc* represents the percentage of solutions of one

Table 1
Limits of parameters of the proposed model.

Parameter	Limit	Parameter	Limit	Parameter	Limit
c_{iklj}	[1200–1600]	F_k	[5e + 8,1e + 9]	r_{2j}	[100,200]
c_{ik}	[8–12]	f_k	[2e + 8,5e + 8]	d_{ik}	[0,2000]
rc_i	[5e + 5,2e + 6]	cap_{ir}	[1e + 6,1.5e + 6]	e_i	[0,1]
r_{1j}	[50,80]	t_i	[0,0.9]	cn_i	[5e + 8;7e + 6]

Table 2
The proposed test problems.

Parameters		Test problem
The potential number of the relief center	The potential number of higher-level center	
80	32	1
90	50	2
105	70	3
120	100	4
140	120	5

Pareto set that is dominated by at least one solution of the other Pareto set.

- *SM* metric: The *Sm* represents how well the non-dominated solutions are dispersed in the objective space. Therefore, a Pareto set with a smaller value of *SM* is expected to present a better set of non-dominated solutions.

Because of the diverse scales of the considered objective functions in the proposed model, all the calculated values of the Performance measures are normalized to the interval [0, 1].

5.3.2. Parameter setting

After extensive offline computational experiments, and using the Taguchi experimental design, parameters of the proposed solution algorithm are set (Byrne and Taguchi, 1987). Since the proposed NSGA-II has four parameters, the $L_9(3^4)$ orthogonal array should be used. The VNDS has three parameters and therefore the $L_9(3^3)$ orthogonal array should be used. Similarly, as the NSGA-II-VNDS has six parameters, the $L_{27}(3^6)$ orthogonal array is used. Finally, the $L_{27}(3^5)$ orthogonal array should be used for the proposed NSGA-II-VNS. Regarding instances with different characteristics (i.e., noise factors), the $L_9(3^3)$ orthogonal array is chosen (Table 3). The average pull-off force (i.e., an average of performance measure, P_{avg}) and the signal-to-noise ratio (*SN*) for each level of design factors are presented in Table 3. After some experiments with different performance measures, the *HV* metric is considered as the main performance measure to tune the parameters (Montgomery, 2012). Table 4 shows the final tuned value of the parameters of the proposed algorithms based on the obtained results of 10 times run of each proposed methods.

5.3.3. Determining the effective order of neighborhood structure selection

The procedure of neighborhood selection is improved in the proposed VNDS. To determining an effective selection order of the neighborhood structures, a fitness landscape analysis approach is used. First, the *Dmm* is calculated for each neighborhood structure via applying a hill-climbing algorithm that incorporates one combination to improve a population. After some experiments with different performance measures, the *HV* metric is considered as the main metric to calculate the *Dmm*. For this matter, the proposed test problems are used for ten times with an initial similar population size of 100. The obtained results of the *Dmm* computation are outlined in Fig. 6. These results indicate that all the applied neighborhood structures have a significant positive impact on concentrating examined populations. The neighborhood structures with a smaller *Dmm* value have a more focused concentration. The order of selection of neighborhood structures is determined based on the ascending order of the *Dmm* values. The results indicate that the neighborhood structures (*K6*, *K9*, *K7*, *K3*, *K8*, *K1*, *K5*, *K4*, and *K2*) are selected sequentially in each iteration of the proposed VNDS.

5.3.4. The efficiency of the proposed hybrid solution algorithm

In this section, the quality of the obtained Pareto optimal solution is assessed. The best known *HV*, HV_{best}, $HV_{average}$, $Setc_{average}$, Sm_{best}, $Sm_{average}$, and the average CPU time are reported based on the ten times runs (see Table 5). Obtained results indicate that the HTNSGAII-VNDS found superior Pareto solutions for all instances according to the closeness of the solutions to the Pareto-optimal set and the spread of the solutions across the objective space. However, in order to strengthen the obtained results, the *Setc* metric is also considered. The results of coverage relationship assessment (Table 6) indicate that the HTNSGAII-VNDS found superior Pareto solutions for all instances. The results of the *Setc* metric assessment indicate that an average 76.7%, 75.6%, and 56.5% of the Pareto solution found by the VNDS, NSGA-II, and NSGA-II-VNS are dominated by solutions found by the NSGA-II-VNDS, respectively, while these meta-heuristics are able to dominate only 1.1%, 9.8%, and 31.7% of the Pareto solution found by the HTNSGAII-VNDS. Similarly, this significant superiority is also observed in the other comparisons. In addition, results of Table 7 indicate that solutions found by the NSGAII-VNDS are better than those found by the other considered meta-heuristics in terms of the *SM* metric. For instance, the average distribution of the non-dominated solutions obtained by the HTNSGAII-VNDS in instance 5 is 32.1% of the maximum normalized distance between the reference point and the origin of coordinates.

Table 3
The considered design and noise factors in the Taguchi design technique.

Type	Factors	Algorithms				Levels		
		VNDS	NSGA-II	NSGA-II-VNS	HTNSGA-II-VNDS	Level1	Level2	Level3
Design	Population size	✓	✓	✓	✓	75	100	150
	Generation number		✓	✓	✓	300	400	500
	Pareto-list size		✓	✓	✓	10	15	20
	Crossover rate		✓	✓	✓	0.7	0.8	0.9
	Neighborhood structure iteration	✓		✓	✓	5	10	15
	Local search iteration	✓		✓	✓	3	4	5
Noise	Number of the relief center	✓	✓	✓	✓	32	70	120
	Number of higher-level centers	✓	✓	✓	✓	80	105	140

Table 4
The final parameter setting of the proposed algorithms.

Algorithms	Population size	Generation number	Pareto-list size	Crossover rate	Neighborhood structure iteration	Local search iteration
VNDS	150	400	20	0.8	15	4
NSGA-II	150	400	20	0.8	15	4
NSGA-II-VNS	100	300	15	0.8	15	4
HTNSGA-II-VNDS	100	300	15	0.8	10	3

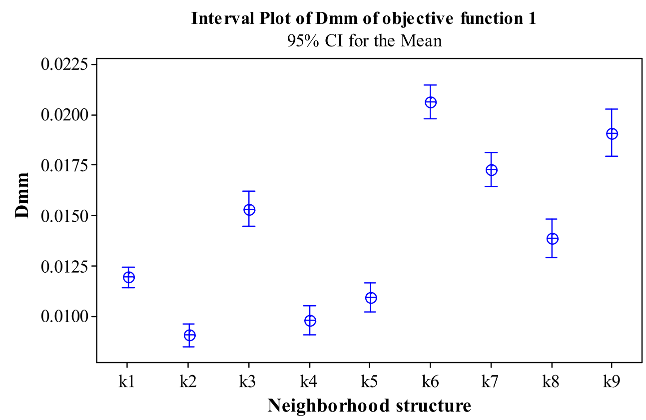

Fig. 6. Interval plot of the calculated *Dmm*.

Mann-Whitney *U* test was done to investigate the statistical significance of the HTNSGAII-VNDS superior performance. The U-value is computed zero for all the possible performance comparisons. Since the critical value of U at $p \le 0.05$ is 2 (based on the number of observations), all the results are significant at $p \le 0.05$. These obtained results ensure the superiority of the HTNSGAII-VNDS against all the other methods. From the CPU time perspective (Table 8), the NSGA-II has better performance than all the other methods. However, this better performance is overshadowed by the fact NSGA-II has found worse solutions than the other solution algorithms.

5.3.5. Sensitivity analysis and managerial insights

In this section, an extensive set of experiments is performed to analyze the capability of the proposed network design model to deal with disasters conditions and response to customers' demands. Fig. 7 depicts the conflict between the considered objective functions of the proposed relief network design. To investigate this conflict, the proposed problem is solved via using three consecutive procedures. First, the problem is solved by considering a single objective, such as the first objective function. The solution that is obtained via optimizing the reduced single objective problem is used for determining the values of the other objective functions. Then, the problem is solved again by considering the other objective functions as new constraints. It should be mentioned that the values of right-hand-side parameters of the new constraints are equal to the last obtained optimal values of these objective functions. This procedure is repeated for the other objective functions. In addition, to investigate the impact of the coverage distance on the total risk of relief network and the covered population under disaster, the proposed network design model is solved for various coverage distances. The obtained results show the decrease in the risk as a consequence of increase in the coverage distance. The effect of changes in the coverage distance on the risk is different for diverse levels of coverage distance. The main reason for this phenomenon comes from a number of variations in the status of the established facilities for relief network designing. The results of Fig. 8 demonstrate

Table 6
Results of solution algorithm efficiency assessment using the *Setc* metric.*

Test problem		VNDS	NSGA-II	NSGA-II-VNS	HTNSGA-II-VNDS	Average
1	VNDS	–	0.02	0.01	0.01	0.01
	NSGA-II	0.38	–	0.17	0.13	0.23
	NSGA-II-VNS	0.59	0.24	–	0.20	0.34
	HTNSGA-II-VNDS	0.68	0.63	0.49	–	0.60
	Average		0.55	0.23	0.11	
2	VNDS	–	0.01	0.02	0.01	0.01
	NSGA-II	0.22	–	0.07	0.08	0.12
	NSGA-II-VNS	0.47	0.34	–	0.32	0.38
	HTNSGA-II-VNDS	0.73	0.73	0.54	–	0.67
	Average		0.47	0.21	0.14	
4	VNDS	–	0.02	0.02	0.01	0.02
	NSGA-II	0.23	–	0.08	0.09	0.13
	NSGA-II-VNS	0.50	0.35	–	0.34	0.39
	HTNSGA-II-VNDS	0.77	0.77	0.58	–	0.71
	Average		0.50	0.23	0.15	
4	VNDS	–	0.02	0.02	0.01	0.02
	NSGA-II	0.23	–	0.08	0.09	0.13
	NSGA-II-VNS	0.51	0.35	–	0.36	0.41
	HTNSGA-II-VNDS	0.80	0.81	0.59	–	0.74
	Average		0.52	0.23	0.15	
5	VNDS	–	0.02	0.02	0.01	0.02
	NSGA-II	0.24	–	0.08	0.10	0.14
	NSGA-II-VNS	0.53	0.37	–	0.37	0.42
	HTNSGA-II-VNDS	0.85	0.84	0.62	–	0.77
	Average		0.54	0.24	0.16	

* 0.00 i.e., 00.00%.

Table 5
Results of solution algorithm efficiency assessment using the *HV* metric.

Test problem	VNDS		NSGA-II		NSGA-II-VNS		HTNSGA-II-VNDS	
	HV_{best}	$HV_{average}$	HV_{best}	$HV_{average}$	HV_{best}	$HV_{average}$	HV_{best}	$HV_{average}$
1	0.716	0.675	0.756	0.730	0.831	0.784	0.884	0.916
2	0.697	0.652	0.741	0.704	0.825	0.789	0.879	0.900
3	0.685	0.642	0.732	0.703	0.822	0.792	0.866	0.896
4	0.658	0.616	0.704	0.661	0.778	0.743	0.829	0.856
5	0.736	0.678	0.785	0.758	0.861	0.836	0.911	0.942

Table 7
Results of solution algorithm efficiency assessment using the *Sm* metric.*

Test problem	VNDS		NSGA-II		NSGA-II-VNS		HTNSGA-II-VNDS		Best known solution (BKS)
	SM_{best}	$SM_{average}$	SM_{best}	$SM_{average}$	SM_{best}	$SM_{average}$	SM_{best}	$SM_{average}$	
1	0.356	0.348	0.326	0.355	0.304	0.295	0.291	0.283	0.291
2	0.464	0.424	0.453	0.443	0.395	0.390	0.369	0.371	0.369
3	0.452	0.434	0.431	0.408	0.388	0.383	0.370	0.364	0.370
4	0.367	0.346	0.324	0.348	0.302	0.293	0.289	0.282	0.289
5	0.386	0.370	0.357	0.376	0.334	0.331	0.321	0.312	0.321

* 0.000 i.e., 00.00%.

Table 8
Average CPU run time of the solution algorithms.

Test problem	VNDS	NSGA-II	NSGA-II-VNS	HTNSGA-II-VNDS	BKS
1	1.86	1.41	2.20	2.33	1.41
2	2.88	2.28	3.32	3.63	2.28
3	4.46	3.91	4.75	4.76	3.91
4	5.70	5.43	5.98	6.06	5.43
5	6.41	5.93	6.94	7.26	5.93

that the value of risk measure declines about 12 percent as a consequence of changing the coverage distance from 200 to 150 km. In addition, the decrease in the value of risk measure is 0.12 percent when the coverage distance is altered from 350 to 200 km. The achieved results show that increase in the coverage distance does not necessarily lead to a decrease in the service risk. This consequence occurs because one relief facility could be located at the similar location and the similar point determines the maximum risk value for those considered coverage distances. Results of investigating the relation between population coverage and distance coverage show that the population coverage does not follow a monotonic pattern like as a risk even though there is an increasing trend. The population coverage increases with increasing coverage distance as expected if the location of relief facility does not change. In addition, the population coverage may grow or decline when the location of facility changes. Fig. 9 represents the decrease in network risk as a consequence of the increase in a number of relief facilities that cover demand points. Finally, considering the variance of the key performance measures provide useful information to decision makers of relief network as they seek to increase network performance and reduce the service risk. Variance analysis looks at the performance of a conducted decision and compares actual performance with standard performance. Management establishes the standards against which performance will be measured, such as a total cost of the relief network management. The more variance of the Γ_s (See Eq. (20)), the more the risky will be decision making. Therefore, the decision maker can consider a weight (i.e., λ) for the robust counterpart of the uncertain model (see Eq. (18)). While considering the variance of the obtained solutions in the proposed model leads to a slight increase in total cost of the relief network management, but improvement in the stability of the proposed plan under uncertainties is more important (see Fig. 10). Stability in resource consumption, such as relief budget, is very valuable due to extreme resource constraints. Because of facing with the high risk of earthquake occurrence, improvement in the efficiency of the resource usage can have a significant impact on the quality of relief service. To examine the performance of the proposed inventory grouping approach over time, we implement the proposed mathematical programming model and classical *ABC* grouping method solutions under considered planning period. Obtained results indicate that when there is a sufficient inventory budget, the optimal number of inventory groups which is determined by the optimization model converges to three. Results show a decrease in the average cost of network management by an increase in a number of inventory groups. Therefore, the quality of the classical *ABC* solution is close to the quality of the inventory grouping optimization model. But the number of inventory groups needs to increase when the inventory budget is tight and the classical *ABS* solution will not be any more efficient (see Fig. 11). In addition, a number of inventory groups have a significant effect on the total coverage and risk of the relief network.

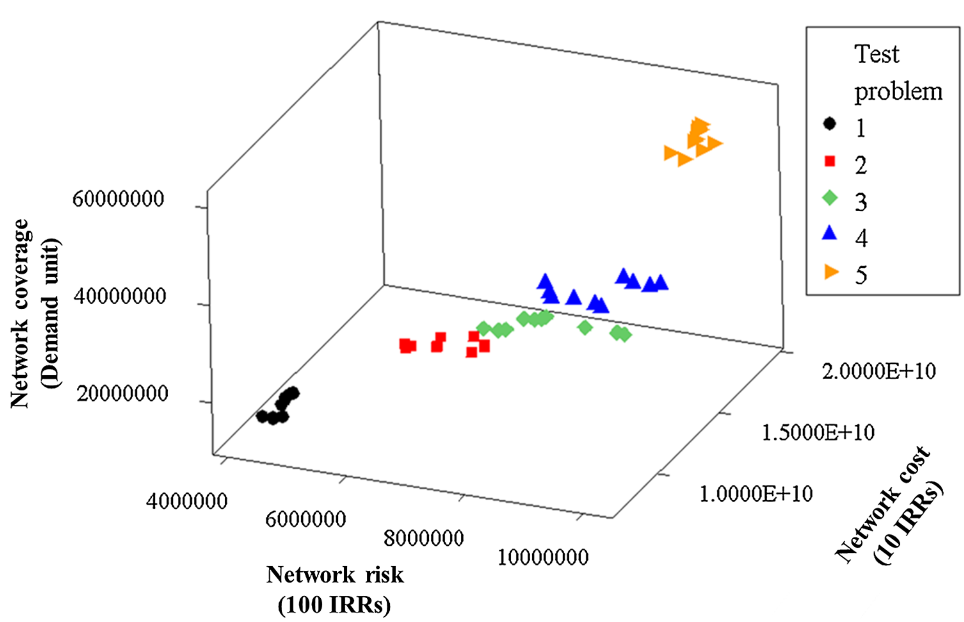

Fig. 7. The conflict between the three objectives of the proposed relief network.

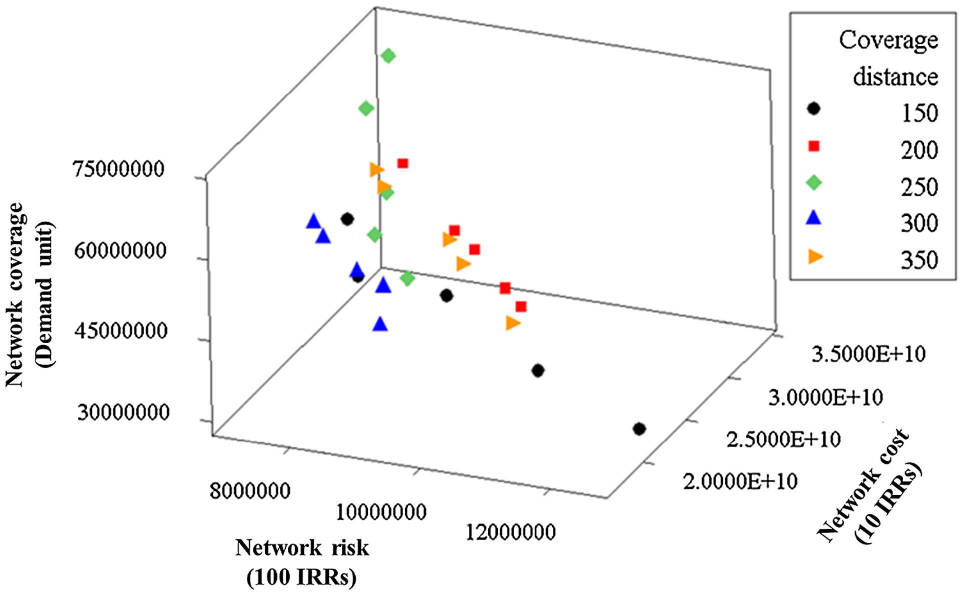

Fig. 8. Impact of change in the coverage distance on relief network performance.

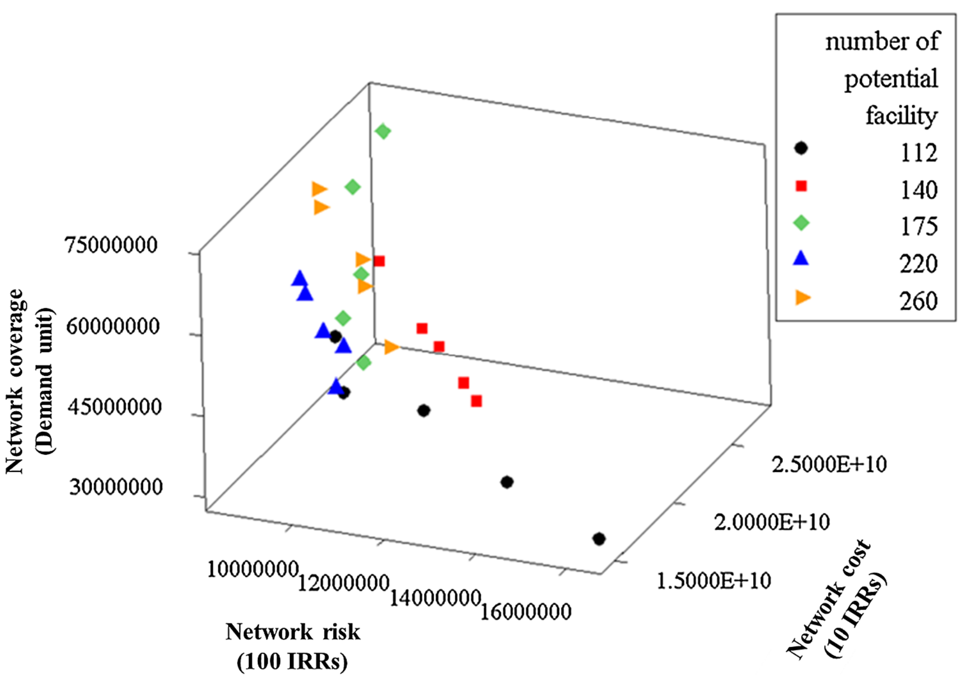

Fig. 9. Impact of increase in the number of potential relief facilities on relief network performance.

6. Conclusion and future research areas

In this study, a scenario-based robust multi-objective relief network design model was proposed for a pre-disaster phase of a disaster management. The vulnerability issue of risk is considered via using the FTA method. The locations of the relief facilities are not already known, but the vulnerability of a demand point is directly dependent on where the relief facilities are established. Therefore, the facility location decisions are to be integrated into the vulnerability computation. In addition to minimizing the service providing risk issue as an objective function of the proposed model, other objective functions include maximizing the network coverage and minimizing total cost of relief network are considered. Due to the high complexity of the large-scale network design problem, a multi-objective evolutionary algorithm incorporating the Taguchi-based NSGA-II and improved VNDS with fitness landscape analysis was designed to solve a proposed model. The efficiency of the proposed algorithm is investigated by comparing its performance with other similar meta-heuristics. The results obtained from this study indicate the efficiency of the proposed modeling approach for designing the relief network via considering disaster management issues. The specific range of a covering distance and a number of established facilities has a major impact on the network risk. The obtained results demonstrate that considering the risk measure in network design could result in significant differences in the risk levels. For further research, various strategic and tactical decisions such as a facility capacity planning, vehicle routing in disaster areas, as well as determining an appropriate inventory policy at relief facilities. While the effects of disruption on facility capacity were considered in this study, other important disruptions such as capacity and availability of transportation routes should be considered as well. In such situations,

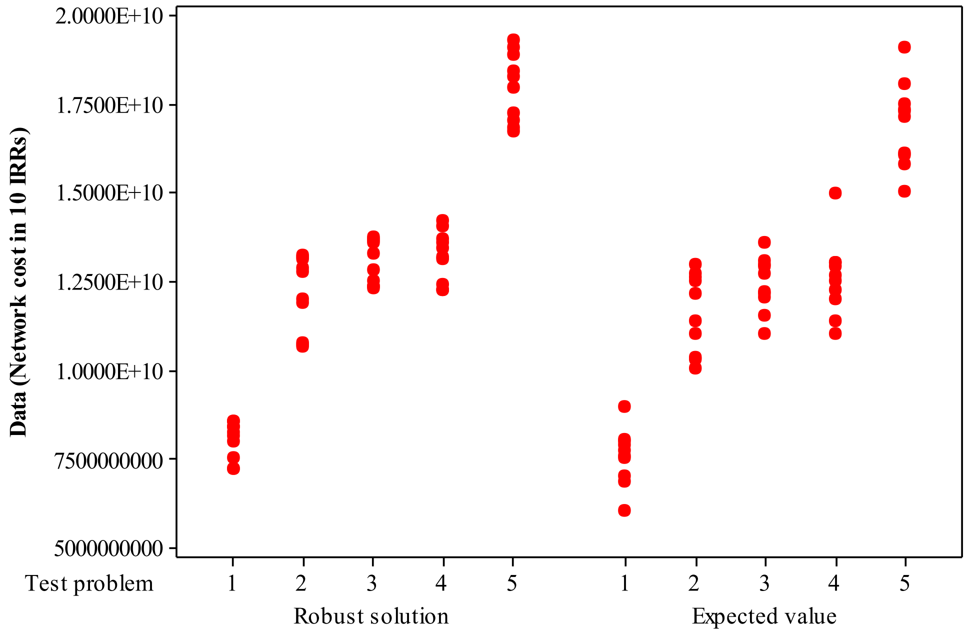

Fig. 10. Individual plot of the obtained robust and stochastic solutions.

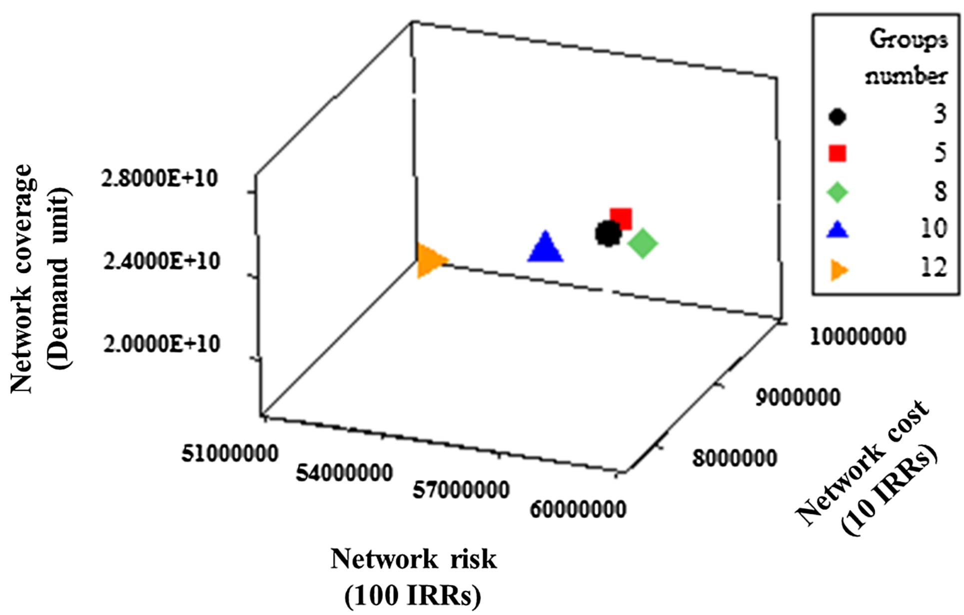

Fig. 11. Impact of increase in the number of inventory groups on relief network performance.

transportation lead times are subject to various risks and uncertainties. Therefore, developing a mathematical model and appropriate solution algorithm to tackle such situations could also be of interest.

Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.ssci.2018.09.004.

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